

Abstract

We consider a quiz show game in which a contestant is presented with a sequence of questions. Each time the contestant answers a question correctly, she receives a prize and proceeds to the next question; the probability of answering each question correctly is given. If the contestant answers a question incorrectly, she receives a consolation prize and the game ends. The contestant's problem of determining the optimal order in which to answer questions, for known fixed parameters, is a classic one studied in Kadane (1969), and admits a simple index-based solution. We consider game-theoretic versions of this problem in which a game show host can choose how to allocate a fixed prize budget. Our models are motivated by operational search problems in national security involving reconnaissance missions and inspecting for evidence of nuclear enrichment, as well as certain scheduling problems. We study three variants of the game, corresponding to different ways in which the host can distribute the prize money. For the first variant, we provide complete closed-form solutions, including equilibrium strategies and the value of the game. We reduce the second variant to a game solved in the literature. For the third variant, we obtain partial results by analyzing a more general game with a geometric structure.

Keywords: search games; zero-sum games; submodularity; defense